

5G RAN

Scalable Bandwidth Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the setup of 30 MHz NR TDD cells in the n38 frequency band. For details, see 4.1.1.1 NR TDD Cells Served by Macro Base Stations .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added support for high-frequency NR TDD by LampSite base stations. For details, see: • 4.1.1.2 NR TDD Cells Served by LampSite Base Stations • 4.3.3 Hardware	None	High-frequency TDD	DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Revised descriptions of the impact relationship between "multi-carrier combination selection based on multiple bandwidth levels" and UE bandwidth adaptation due to the change of the parameter for controlling "multi-carrier combination selection based on multiple bandwidth levels". For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Added descriptions of the impact relationship between UE bandwidth adaptation and experience-based smart carrier selection. For details, see [5.3.2.3 Function Impacts](#).
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010007	Scalable Bandwidth	4 Cell Bandwidth 5 UE Bandwidth Adaptation

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Cell bandwidth	None	4 Cell Bandwidth
UE bandwidth adaptation	None	5 UE Bandwidth Adaptation

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Cell bandwidth	Supported in both low and high frequency bands, with the following differences: Different cell bandwidths are used in high and low frequency bands. For details, see 4.1 Principles .	4 Cell Bandwidth
UE bandwidth adaptation	Supported only in low frequency bands	5 UE Bandwidth Adaptation

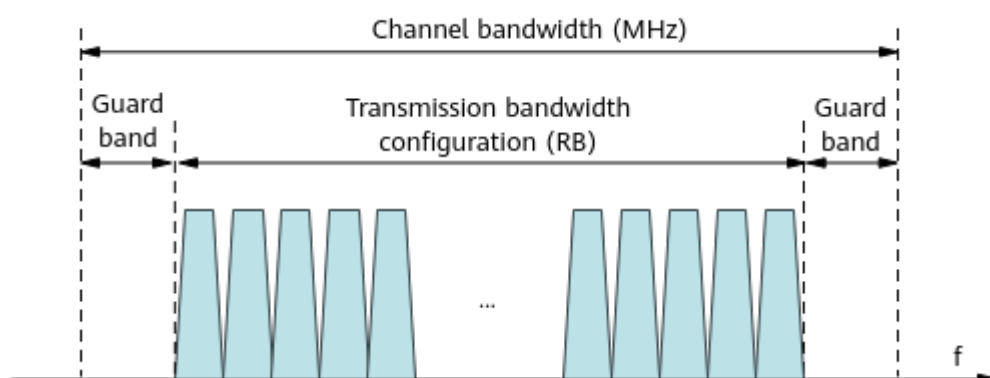
3 Overview

NR spectrum resources consist of several frequency bands which are numbered in the format of "n+integer" and divided based on spectrum usage and future planning. There may be issues such as inconsistent bandwidths and spectrum range overlapping between these bands.

As shown in [Figure 3-1](#), a channel bandwidth (in MHz) consists of a transmission bandwidth configuration (in RBs) and guard bands (in kHz) on both ends. The transmission bandwidth configuration is used for data transmission, while the guard bands help isolate interference by bands beyond the channel bandwidth during data transmission.

A cell bandwidth refers to the channel bandwidth occupied by a cell.

Figure 3-1 Channel bandwidth



- The standard cell bandwidth is defined in section 5.3.5 "BS channel bandwidth per operating band" in 3GPP TS 38.104 V15.5.0. Different frequency bands support different cell bandwidths and subcarrier spacings. [4 Cell Bandwidth](#) describes the cell bandwidths supported by Huawei macro and LampSite base stations.
- During the access to a cell, a UE must have the same transmission bandwidth as that configured on the gNodeB. The UE cannot access a cell if the UE-supported bandwidth is lower than the cell bandwidth. To solve this problem,

UE bandwidth adaptation is introduced. For details, see [5 UE Bandwidth Adaptation](#).

4 Cell Bandwidth

4.1 Principles

The **NRDUCell.UlBandwidth**, **NRDUCell.DlBandwidth**, and **NRDUCell.SubcarrierSpacing** parameters can be used to configure the uplink cell bandwidth, downlink cell bandwidth, and subcarrier spacing of the cell, respectively, according to the operator's requirements. Basic cell information must be configured by network planning engineers based on scenarios.

Table 4-1 lists the duplex modes (controlled by the **NRDUCell.DuplexMode** parameter) supported by Huawei base stations. The bandwidths supported by cells in different duplex modes are provided in tables in subsequent subsections. In the tables, "√" indicates that the frequency band supports the bandwidth, and "-" indicates that the frequency band does not support the bandwidth. For details about the subcarrier spacing supported by cells in different duplex modes, see *Cell Management*.

Table 4-1 Duplex modes supported by Huawei base stations

Base Station Type	NR TDD Low-Frequency Band	NR TDD High-Frequency Band	NR SUL
Macro base station	√	√	√
LampSite base station	√	-	-

 NOTE

Unless otherwise stated, bandwidths in **4 Cell Bandwidth** refer to standard bandwidths, which are the cell bandwidths defined in section 5.3.5 "BS channel bandwidth per operating band" in 3GPP TS 38.104 V16.8.0.

4.1.1 Bandwidths Supported by NR TDD Cells

This section describes the bandwidths supported by Huawei NR TDD cells.

4.1.1.1 NR TDD Cells Served by Macro Base Stations

NR TDD Low-Frequency Band

Table 4-2 lists the bandwidths supported by macro base stations in NR TDD low-frequency bands.

Table 4-2 Bandwidths supported by macro base stations in NR TDD low-frequency bands

Cell Bandwidth	n38	n40	n41	n77	n78	n79
20 MHz	√	√	√	√	√	-
30 MHz	√	√	√	√	√	-
40 MHz	√	√	√	√	√	√
50 MHz	-	√	√	√	√	√
60 MHz	-	√	√	√	√	√
70 MHz	-	√	√	√	√	-
80 MHz	-	√	√	√	√	√
90 MHz	-	√	√	√	√	-
100 MHz	-	√	√	√	√	√

NR TDD High-Frequency Band

Table 4-3 lists the bandwidths supported by macro base stations in NR TDD high-frequency bands.

Table 4-3 Bandwidths supported by macro base stations in NR TDD high-frequency bands

Cell Bandwidth	n257	n258	n261
100 MHz	√	√	√
200 MHz	√	√	√

4.1.1.2 NR TDD Cells Served by LampSite Base Stations

NR TDD Low-Frequency Band

Table 4-4 lists the bandwidths supported by LampSite base stations in NR TDD low-frequency bands.

Table 4-4 Bandwidths supported by LampSite base stations in NR TDD low-frequency bands

Cell Bandwidth	n38	n40	n41	n77	n78
20 MHz	√	√	√	√	√
30 MHz	-	√	√	√	√
40 MHz	√	√	√	√	√
50 MHz	-	√	√	√	√
60 MHz	-	√	√	√	√
70 MHz	-	√	√	√	√
80 MHz	-	√	√	√	√
90 MHz	-	√	√	√	√
100 MHz	-	√	√	√	√

NR TDD High-Frequency Band

Table 4-5 lists the bandwidths supported by LampSite base stations in NR TDD high-frequency bands.

Table 4-5 Bandwidths supported by LampSite base stations in NR TDD high-frequency bands

Cell Bandwidth	n258
100 MHz	√
200 MHz	√

4.1.2 Bandwidths Supported by NR SUL Cells

For details about the bandwidths supported by macro base stations in NR SUL low-frequency bands, see *UL and DL Decoupling* and *Super Uplink*.

4.2 Network Analysis

4.2.1 Benefits

With given channel conditions and hardware, including central processing unit (CPU), digital signal processor (DSP), and memory, cells with a larger bandwidth serve and schedule more UEs, and deliver larger throughput and more satisfactory KPIs. However, these cells occupy more hardware resources.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Changing the bandwidth of an activated NR cell will cause the cell to be automatically reestablished, which affects admitted UEs. For example, the service drop occurs.

If the cell bandwidth becomes larger after being reconstructed, the PSD may decrease when the current RF module works at full power or the cell power is proactively reduced. As a result, the network coverage shrinks while the downlink coverage, UE-perceived throughput, number of online UEs, and other indicators fluctuate.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	None	None	None	None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. The NR-capable cell bandwidths are subject to the capabilities of baseband processing units. For details about NR-capable main control boards and baseband processing units and about the bandwidths supported by baseband processing units, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules can be used, with the supported cell bandwidths depending on the RF module capabilities. For details about NR-capable RF modules and the cell bandwidths they support, see the technical specifications of the related RF modules in *3900 & 5900 Series Base Station Product Documentation*.

LampSite high-frequency NR TDD cells with a 100 MHz or 200 MHz bandwidth can be served only by the pRRU5551JR.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This basic function takes effect by default and does not require activation. [Table 4-6](#) describes the parameters used for function optimization.

NOTE

Modifying the following parameters will interrupt services. Before the modification, refer to "Caution" for the corresponding parameters in the parameter reference.

Table 4-6 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Duplex Mode	NRDUCell.Duplex Mode	None	Set this parameter based on the network plan.
Uplink Bandwidth	NRDUCell.UlBand width	None	Set this parameter based on site requirements.
Downlink Bandwidth	NRDUCell.DlBand width	None	Set this parameter based on site requirements.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

MML commands are not required for activating this function.

Optimization Command Examples

```
//Modifying the uplink and downlink bandwidths of an NR TDD cell  
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, UlBandwidth=CELL_BW_100M,  
DlBandwidth=CELL_BW_100M;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Step 1** Run the **LST NRDUCELL** command to check the static parameter settings of a cell. In the command output, if the displayed uplink and downlink bandwidths are consistent with the actual configurations, go to [Step 2](#).
- Step 2** Run the **DSP NRDUCELL** command. If the value of **NR DU Cell State** is **Normal**, the cell bandwidth is successfully changed.

----End

4.4.3 Network Monitoring

None

5 UE Bandwidth Adaptation

5.1 Principles

In order for a UE to access a cell, the UE must support the cell bandwidth. For example, when the cell bandwidth is 20 MHz, the UE can access the cell only if the UE supports this bandwidth. However, a UE does not support certain cell bandwidths, such as 100 MHz, which is defined as the maximum bandwidth for a low-frequency cell in section 5.3.5 "BS channel bandwidth per operating band" in 3GPP TS 38.104 V16.8.0. In this case, the UE will be unable to access the cell due to non-support for the cell's bandwidth. To address this issue, UE bandwidth adaptation is introduced. This function allows a UE to access a cell even when the bandwidth supported by the UE is lower than the cell bandwidth. For example, with this function, a UE that supports a bandwidth of lower than 100 MHz can also access a 100 MHz cell.

Below is a summary of the subsections of this section:

- **5.1.1 UE-supported Bandwidths** describes how the gNodeB obtains UE-supported bandwidths.
- **5.1.2 UE Bandwidth Adaptation** describes UE bandwidth adaptation. After the UE-supported bandwidths are obtained, the UE that does not support the cell bandwidth can access the cell on a BWP if UE bandwidth adaptation is enabled and the BWP is configured.
- **5.1.3 UE Bandwidth Adaptation Optimization** describes optimizations on UE bandwidth adaptation, which include uplink signal leakage interference optimization and PUCCH frequency-domain position optimization.

5.1.1 UE-supported Bandwidths

In a 90 MHz cell: The UE-supported uplink and downlink bandwidths are indicated by the following information elements (IEs) in the UECapabilityInformation message:

- If the value of the **channelBW-90mhz** field in the *featureSetsUplinkPerCC* IE is **supported**, the UE supports the uplink bandwidth of 90 MHz.
- If the value of the **channelBW-90mhz** field in the *featureSetsDownlinkPerCC* IE is **supported**, the UE supports the downlink bandwidth of 90 MHz.

- In other cases, the UE does not support 90 MHz.

In a non-90 MHz cell: The gNodeB allows that a UE reports two types of its bandwidth capabilities through the UECapabilityInformation message.

- UE-supported bandwidths in the serving frequency band are listed in [Table 5-1](#). For details, see section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V16.3.0.

Table 5-1 UE-supported bandwidth capabilities in the serving frequency band

Cell Bandwidth	UE-supported Bandwidth in the Serving Frequency Band
5 MHz, 10 MHz, 15 MHz, 20 MHz, 25 MHz, 30 MHz, 40 MHz, 50 MHz, 60 MHz, 80 MHz	If a UE does not report the channelBWs-UL or channelBWs-DL field, the uplink or downlink bandwidths supported by the UE in the serving frequency band are as specified in Table 5.3.5-1 Channel bandwidths for each NR band in 3GPP TS 38.101 V15.5.0. If a UE reports the channelBWs-UL or channelBWs-DL field, the bandwidths supported by the UE in the serving frequency band are indicated by the 10 bits of the corresponding binary bit string for the field. The 10-bit binary number indicates support for the cell bandwidths listed in the left column. The first and second bits starting from the left indicate support for 5 MHz and 10 MHz, respectively, and this rule also applies to the remaining bits. • If the value of a bit is 1, the UE supports the corresponding bandwidth. • If the value of a bit is 0, the UE does not support the corresponding bandwidth.
70 MHz	The uplink and downlink bandwidths supported by the UE in the serving frequency band are indicated by the leftmost bit of the corresponding binary bit string for the channelBWs-UL-v1590 and channelBWs-DL-v1590 fields, respectively. • If the value is 1, the UE supports the corresponding bandwidth. • If the value is 0 or this field does not exist, the UE does not support the corresponding bandwidth.

Cell Bandwidth	UE-supported Bandwidth in the Serving Frequency Band
100 MHz	In the n40 frequency band: The uplink and downlink bandwidths supported by the UE are indicated by the fourth leftmost bit of the corresponding binary bit string in the channelBWs-UL-v1590 and channelBWs-DL-v1590 fields, respectively. • If the value is 1, the UE supports the 100 MHz bandwidth. • If the value is 0 or this field does not exist, the UE does not support the 100 MHz bandwidth. In other frequency bands: Whether the 100 MHz bandwidth is supported is checked according to "Table 5.3.5-1 Channel bandwidths for each NR band" in 3GPP TS 38.101 V15.5.0. • If it is supported, the UE supports the 100 MHz bandwidth. • If it is not supported, the UE does not support the 100 MHz bandwidth.

- UE-supported maximum bandwidths in the serving cell: The UE-supported maximum uplink and downlink bandwidths in the serving cell are indicated by the following IEs, respectively:
 - The **supportedBandwidthUL** or **supportedBandwidthUL-v1710** field in the *featureSetsUplinkPerCC* IE indicates the UE-supported maximum uplink bandwidth.
 - The **supportedBandwidthDL** or **supportedBandwidthDL-v1710** field in the *featureSetsDownlinkPerCC* IE indicates the UE-supported maximum downlink bandwidth.

For details, see sections 4.2.7.6 "FeatureSetDownlinkPerCC parameters" and 4.2.7.8 "FeatureSetUplinkPerCC parameters" in 3GPP TS 38.306 V15.5.0.

Assume that the subcarrier spacing is 30 kHz. If a UE reports **0001001111** in the **channelBWs-DL-v1530** field (indicating the support for downlink bandwidths of 20 MHz, 40 MHz, 50 MHz, 60 MHz, and 80 MHz) and also reports the support for a maximum downlink bandwidth of 60 MHz in the **supportedBandwidthDL** field, the UE supports downlink cell bandwidths of 20 MHz, 40 MHz, 50 MHz, and 60 MHz.

NOTE

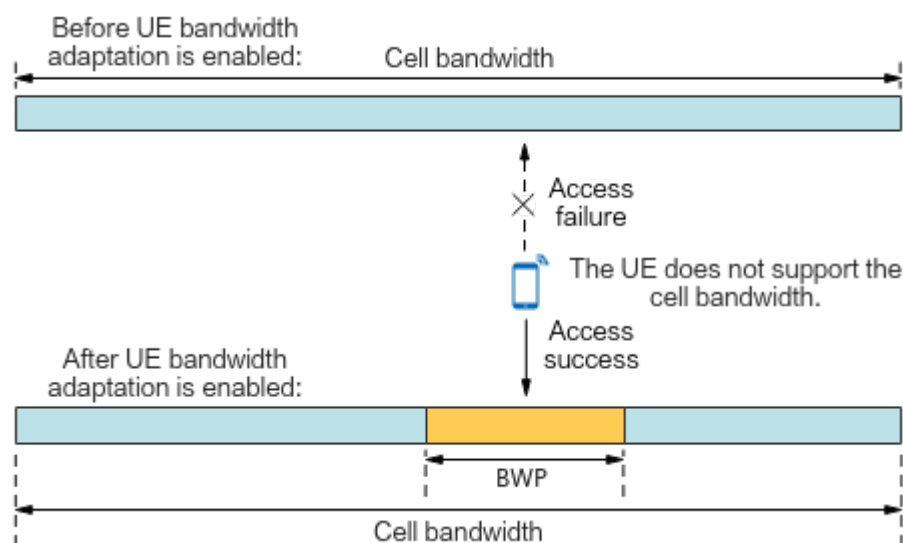
- For the n40 frequency band, the **CHANNEL_BW_100M_UE_CAP_R16_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter specifies whether gNodeBs use the **channelBWs-DL-v1590** and **channelBWs-UL-v1590** fields to determine the 100 MHz bandwidth capability of a UE. This option is selected by default. If a UE reports the two capability fields, the base station uses these fields to determine whether the UE supports the 100 MHz bandwidth. To support the access of UEs that do not support 100 MHz bandwidth in 3GPP Release 15, it is recommended that UE bandwidth adaptation be enabled and the **NRDUCellBwp.UIMinCarrierBw** and **NRDUCellBwp.DIMinCarrierBw** parameters be set to 80 MHz.
- In TDD scenarios, UEs supporting both the uplink and downlink cell bandwidths are considered to support the corresponding cell bandwidth.

- In SUL scenarios, UEs supporting the uplink cell bandwidth are considered to support the corresponding cell bandwidth.

5.1.2 UE Bandwidth Adaptation

After UE-supported bandwidths are obtained, the UE will fail to access the cell if the UE-supported bandwidths do not match the cell bandwidth. In this case, UE bandwidth adaptation can be enabled and a segment of bandwidth (that is, BWP) that matches the UE-supported bandwidths can be configured for the UE. Then, the UE can access the cell through the BWP. [Figure 5-1](#) shows the principles of UE bandwidth adaptation.

Figure 5-1 Principles of UE bandwidth adaptation



UE bandwidth adaptation is enabled by using the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp.BwpConfigSwitch** parameter. If this option is selected, the gNodeB checks whether the UE supports the serving cell's bandwidth.

- Supported: The cell bandwidths specified by **NRDUCell.UlBandwidth** and **NRDUCell.DlBandwidth** take effect for the UE.
- Not supported: The minimum carrier bandwidths specified by **NRDUCellBwp.UlMinCarrierBw** and **NRDUCellBwp.DlMinCarrierBw** take effect for the UE.

[Table 5-2](#) lists the bandwidth configuration requirements in different duplex modes.

Table 5-2 Bandwidth configuration requirements in different duplex modes

Duplex Mode	Configuration Requirement
NR TDD	NRDUCellBwp.UlMinCarrierBw and NRDUCellBwp.DlMinCarrierBw can be set to values within the value ranges and less than the cell bandwidths (except for NOT_CONFIG). The two parameters must be set to the same value.

Duplex Mode	Configuration Requirement
NR SUL	NRDUCellBwp.UlMinCarrierBw can be set to a value within the value range and less than the cell bandwidth (except for NOT_CONFIG). NRDUCellBwp.DlMinCarrierBw does not need to be configured and the default value is NOT_CONFIG .

 NOTE

For a 90 MHz cell:

- When UE bandwidth adaptation is disabled, all UEs that support either the 90 MHz or 80 MHz cell bandwidth can access the cell.
- When UE bandwidth adaptation is enabled, the cell bandwidth is configured for the UEs which support the 90 MHz cell bandwidth. If the UEs do not support the 90 MHz cell bandwidth, the UE-level bandwidths with the same values as **NRDUCellBwp.UlMinCarrierBw** and **NRDUCellBwp.DlMinCarrierBw** are configured for the UEs in addition to the 90 MHz cell bandwidth.

5.1.3 UE Bandwidth Adaptation Optimization

After UE bandwidth adaptation takes effect, uplink signal leakage interference optimization and PUCCH frequency-domain position optimization can also be enabled to prevent uplink signal leakage and uplink throughput decrease.

Uplink Signal Leakage Interference Optimization

After UE bandwidth adaptation takes effect, the UE-allowable uplink bandwidth is smaller than the cell bandwidth. In this case, the guard bandwidth corresponding to the UE-allowable uplink bandwidth may be greater than the guard bandwidth corresponding to the cell bandwidth. For details, see "Table 5.3.3-1: Minimum guardband for each UE channel bandwidth and SCS (kHz)" in 3GPP TS 38.101 V15.5.0. For example, the UE-allowable uplink minimum carrier bandwidth (specified by **NRDUCellBwp.UlMinCarrierBw**) is 80 MHz, and the bandwidth of the serving cell (specified by **NRDUCell.UlBandwidth**) is 100 MHz. As stipulated in "Table 5.3.3-1: Minimum guardband for each UE channel bandwidth and SCS (kHz)" in 3GPP TS 38.101, when the subcarrier spacing is 30 kHz, the 80 MHz bandwidth corresponds to a guard bandwidth of 925 kHz and the 100 MHz bandwidth corresponds to a guard bandwidth of 845 kHz. In this case, the UE may experience the uplink signal leakage of 80 kHz (925 kHz – 845 kHz = 80 kHz).

In the preceding scenarios, the **UL_SIGNAL_LEAK_OPT_SW** option of the **NRDUCellBwp.BwpConfigSwitch** parameter can be selected to enable the gNodeB to puncture the uplink BWP bandwidth of the UE, preventing uplink signal leakage.

PUCCH Frequency-Domain Position Optimization

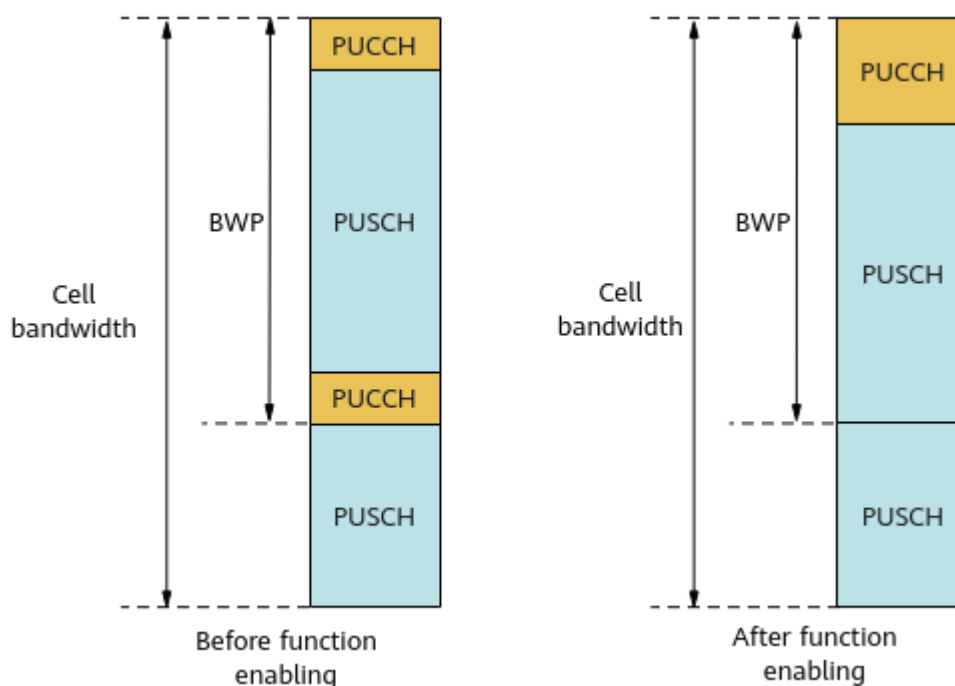
After UE bandwidth adaptation takes effect, there may be two types of UEs in the cell:

- If the UE supports the serving cell's bandwidth, the cell bandwidth takes effect for the UE.
- If the UE does not support the serving cell's bandwidth, the minimum carrier bandwidth takes effect for the UE.

To maximize the reuse of PUCCH resources, the PUCCH is configured within the overlapping scope of these bandwidths and is not configured at both ends of the frequency-domain position of the cell. As a result, RBs of the PUSCH are separated by those of the PUCCH, which may decrease the uplink throughput.

In the preceding scenario, observe the values of **N.UL.NI.Avg.PRBO** to **N.UL.NI.Avg.PR272**. When the counters related to PUCCH RBs are normal, meaning that these RBs do not experience strong interference, the **PUCCH_FREQ_POS_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter can be selected to enable PUCCH frequency-domain position optimization. In this case, the gNodeB configures the PUCCH only at one end of the overlapping scope of these bandwidths, thereby preventing a decrease in uplink throughput. **Figure 5-2** shows PUCCH frequency-domain position optimization.

Figure 5-2 PUCCH frequency-domain position optimization



5.2 Network Analysis

5.2.1 Benefits

When UEs with different allowable bandwidth capabilities need to access a cell, the UE cell bandwidth adaption function can be enabled to provide the following benefits:

- Increases the number of cell access attempts (only in SA networking) and the cell access success rate (in both SA and NSA networking).
- Increases the number of served UEs, traffic volume, and PRB usage in a cell.

Uplink signal leakage interference optimization reduces the risk of uplink signal leakage of a UE when the guard bandwidth of **NRDUCellBwp.UlMinCarrierBw** exceeds the guard bandwidth of **NRDUCell.UlBandwidth**.

PUCCH frequency-domain position optimization improves uplink throughput by optimizing the PUCCH frequency-domain position.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

Table 5-3 lists the impacts of functions on the network.

Table 5-3 Network Impacts

Function Name	Network Impacts
UE bandwidth adaptation	• UE-supported bandwidths of some UEs in the cell are lower than the cell bandwidth. As a result, the cell and UE throughputs decrease. • The actually available PDCCH bandwidth of a cell cannot exceed NRDUCellBwp.DlMinCarrierBw. As a result, the number of available PDCCH symbols may increase and the number of PDSCH symbols may decrease, thereby decreasing the peak cell and UE throughputs. In addition, the number of available PDCCH CCEs in the cell may decrease, which in turn decreases the number of actually used PDCCH CCEs. In heavy-load scenarios, cell PDCCH resource congestion is more likely to occur. • If UE bandwidth adaptation is mistakenly enabled in a cell serving UEs with only one bandwidth capability, the traffic volume and PRB usage will decrease.
Uplink signal leakage interference optimization	The number of available uplink RBs, uplink throughput, and uplink PRB usage decrease.
PUCCH frequency-domain position optimization	When the PUCCH RBs experience strong uplink interference, the uplink coverage of the PUCCH shrinks.

5.3 Requirements

5.3.1 Licenses

There are no license requirements for basic functions.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Full-bandwidth initial BWP configuration	INIT_BWP_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicySwitch parameter	<i>5G Networking and Signaling</i>	The UE bandwidth adaptation function depends on the full-bandwidth initial BWP configuration.
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	Before enabling uplink signal leakage interference optimization (by selecting the UL_SIGNAL_LEAK_OPT_SW option of the NRDUCellBwp.BwpConfigSwitch parameter), ensure that UE bandwidth adaptation has been enabled (by selecting the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter) and the NRDUCellBwp.UlMinCarrierBw parameter is not set to NOT_CONFIG .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	- When the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter is selected, the NRDUCellBwp.UIMinCarrierBw and NRDUCellBwp.DIMinCarrierBw parameters cannot be both set to NOT_CONFIG. - When the NRDUCellBwp.UIMinCarrierBw or NRDUCellBwp.DIMinCarrierBw parameter is set to NOT_CONFIG, the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter must be selected. - When the PUCCH_FREQ_POS_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter is selected, the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter must be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation for multiple BWP1s	PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	When the NRDUCellBwp.DlMinCarrierBw parameter is set to NOT_CONFIG and the REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter is deselected, PDCCH uplink-to-downlink CCE ratio adaptation for multiple BWP1s must be disabled.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	80 MHz UE access	80M_UE_ACCESS_SW option of the NRDUCellUeCoop.UeAccessSwitch parameter	None	After the 80 MHz UE access function is enabled, the effective bandwidth for a UE whose allowable bandwidth capabilities do not allow for 100 MHz in the 100 MHz cell running on the n41 frequency band ranges from 20 MHz to 100 MHz. After UE bandwidth adaptation is enabled, the UE effective bandwidth range cannot be determined, which affects UE access. Therefore, UE bandwidth adaptation and 80 MHz UE access cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Full-bandwidth BWPO configuration upon LTE and NR spectrum sharing	LNR_SPCT_SHR_BWP0_FULL_BW_SW option of the NRDUCellAlgoSwitch.BwpConfigPolicySwitch parameter	None	Full-bandwidth BWPO configuration upon LTE and NR spectrum sharing cannot be enabled when UE bandwidth adaptation is enabled in an NR TDD cell.
Low-frequency TDD	Staggered PRACH frequency-domain resource allocation	PRACH_FREQ_POS_STAGGER_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter	<i>Random Access Control</i>	Staggered PRACH frequency-domain resource allocation cannot work with PUCCH frequency-domain position optimization (controlled by the PUCCH_FREQ_POS_OPT_SW option of the NRDUCellPrach.PrachAlgoExtSwitch parameter).
Low-frequency TDD	Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3	FORMAT1_FREQ_INTF_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	Frequency-domain resource staggering for PUCCH format 1 based on PCI mod 3 (controlled by the FORMAT1_FREQ_INTF_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter) is not supported when all the following conditions are met: • The UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter is deselected. • The NRDUCell.UlBandwidth parameter is set to CELL_BW_30M or CELL_BW_70M.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multiple BWP1s for RedCap	NRDUCellRedCap.RedCapUeMaxAvailableBandwidth set to a value other than 20M	<i>RedCap</i>	- When UE bandwidth adaptation is enabled, the value of the NRDUCellRedCap.RedCapUeMaxAvailableBandwidth parameter must be less than or equal to that of the NRDUCellBwp.UlMinCarrierBw parameter. - When UE bandwidth adaptation is enabled and the NRDUCellBwp.UlMinCarrierBw parameter is set to a value less than 60M, the NRDUCellRedCap.RedCapUeMaxAvailableBandwidth parameter can only be set to 20M.
Low-frequency TDD	SRS antenna selection for RedCap users in FWA scenarios	REDCAP_SRS_ANT_SWITCH_SW option of the NRDUCellRedCap.RedCapAlgoSwitch parameter	<i>FWA</i>	SRS antenna selection for RedCap users in FWA scenarios can be enabled only when either of the following conditions is met: - UE bandwidth adaptation is disabled. - UE bandwidth adaptation is enabled and NRDUCellBwp.DlMinCarrierBw is set to 20M.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH multi-symbol resource pool	MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	When the NRDUCellBwp.DlMinCarrierBw parameter is set to a value other than 80M , 90M , or NOT_CONFIG , the PDCCH multi-symbol resource pool (controlled by the MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) cannot be enabled together with UE bandwidth adaptation.
Low-frequency TDD	Common PUCCH RB resource optimization	COMM_PUCCH_RB_RES_OPT_SW option of the NRDUCellPucch.PucchAlgoSwitch parameter	None	When the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter is selected in a low-frequency TDD cell, the COMM_PUCCH_RB_RES_OPT_SW option can only be deselected.
Low-frequency TDD	PLLC primary cell	PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSwitch.MultiCarrierSwitch parameter	None	When the PLLC_PRIMARY_CELL_SW option of the NRDUCellFeatureSw.MultiCarrierSwitch parameter is selected, the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter must be deselected for the cell.
High-frequency TDD	None	None	None	None

5.3.2.3 Function Impacts

- Functions related to physical channels

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH channel management	NRDUCellPucch.<i>StructureType</i>	<i>Channel Management</i>	When UE bandwidth adaptation is enabled, cells support only long PUCCH time-frequency formats (for which NRDUCellPucch.<i>StructureType</i> is set to LONG_STRUCTURE).
Low-frequency TDD	PUCCH channel management	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.<i>PucchAlgoSwitch</i> parameter	<i>Channel Management</i>	When UE bandwidth adaptation is enabled in TDD cells, NRDUCellBwp.<i>ULMinCarrierBw</i> is set to 20M , and the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.<i>PucchAlgoSwitch</i> parameter is deselected, the sum of NRDUCellPucch.<i>Format1RbNum</i> , NRDUCellPucch.<i>CsiDedicatedRbNum</i> , NRDUCellPucch.<i>Format3RbNum</i> , NRDUCellPucch.<i>Format4RbNum</i> , and NRDUCellPucch.<i>Format4CsiDedicatedRbNum</i> cannot exceed 32.
Low-frequency TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i> parameter	<i>MIMO (TDD)</i>	If UE bandwidth adaptation and CCE resource optimization are both enabled, the latter may not take effect. In this case, it is recommended that the PDCCH_MULTI_BWP1_ALGO_SW option of the NRDUCellPdcch.<i>PdcchAlgoEnhSwitch</i> parameter be also selected.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Narrowband interference	NRDUCellIntrflident.IntrflidentIdentificationMethod and NRDUCellIntrflident.IntrfOptimizationMethod	None	Narrowband interference takes effect only when UE bandwidth adaptation is enabled in TDD cells and NRDUCellBwp.UlMinCarrierBw is greater than or equal to 40M.
Low-frequency TDD	Power saving BWP (low-frequency TDD)	NRDUCellUePwrSaving.BwpPwrSavingSw	<i>UE Power Saving</i>	When enabled, the UE bandwidth adaptation function confines the frequency-domain position of BWP2 (providing a narrow bandwidth) to be within the bandwidth range specified by NRDUCellBwp.UlMinCarrierBw and NRDUCellBwp.DlMinCarrierBw , and adjusts the bandwidth of BWP2 as required.
Low-frequency TDD	Maximum bandwidth reduction	UE_BW_REDUCTION_SW and OVERHEATING_ASSISTANCE_INFO_SW options of the NRCCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	When a UE requests maximum-bandwidth reduction in single-carrier scenarios, if BWP2 is not configured for the UE and the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter is selected, the base station configures uplink/downlink BWPs with the bandwidths specified by NRDUCellBwp.UlMinCarrierBw/ NRDUCellBwp.DlMinCarrierBw for the UE through an RRC reconfiguration message.
Low-frequency TDD	FWA RedCap downlink type 1 scheduling	FWA_TYPE1_4_RB_SCH_SW option of the NRDUCellRedCap.RedCapAlgorithmSwitch parameter	<i>RedCap</i>	In a single-user peak rate test, when both UE bandwidth adaptation and FWA RedCap downlink type 1 scheduling are enabled, the average downlink throughput of RedCap UEs and the average downlink throughput of all UEs in the cell decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_FR1 TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-FR CA does not take effect for UEs with UE bandwidth adaptation in effect.
Low-frequency TDD	Multi-carrier combination selection based on multiple bandwidth levels	MULTI_CARR_ MULTI_BWP_O PT_SW option of the NRCeCellCaMgmtConfig.CaStrategySwitch parameter	<i>Carrier Aggregation</i>	Assume that UE bandwidth adaptation is enabled and minimum carrier bandwidths (NRDUCellBwp.DlMinCarrierBw and NRDUCellBwp.UlMinCarrierBw) are configured. After "multi-carrier combination selection based on multiple bandwidth levels" takes effect, if the bandwidth of the PCell in the optimal combination selected based on the CA capability combination policy differs from the bandwidth in use for a UE, the gNodeB instructs the UE to use the optimal bandwidth.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_ SMART_SEL_SW option of the NRCeCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	After UE bandwidth adaptation is enabled, experience-based smart carrier selection cannot be triggered for a UE identified as an uplink-preferred UE if the effective uplink bandwidth for the UE is not the full uplink bandwidth of the cell.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. The NR-capable cell bandwidths are subject to the capabilities of baseband processing units. For details about NR-capable main control boards and baseband processing units and about the bandwidths supported by baseband processing units, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

In NR TDD, this function requires that the cell bandwidth be greater than 20 MHz.

For an NR TDD cell with **NRDUCell.DlBandwidth** set to **CELL_BW_90M**, if the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp.BwpConfigSwitch** parameter is deselected, the settings of **NRDUCell.SsbFreqPos** for the cell must meet the following requirements:

- If the number of RBs for common control resources is set to **48RB** or **96RB**, the following requirement must be met: $38 \leq \text{Rounddown}[(\text{Frequency associated with the SSB frequency-domain position} - \text{Frequency associated with the cell's downlink NARFCN} + 6 \times N_{\text{RB}} \times \text{SCS}) / (12 \times \text{SCS})] \leq N_{\text{RB}} - 38$.
- If the number of RBs for common control resources is set to **24RB**, the following requirement must be met: $26 \leq \text{Rounddown}[(\text{Frequency associated with the SSB frequency-domain position} - \text{Frequency associated with the cell's downlink NARFCN} + 6 \times N_{\text{RB}} \times \text{SCS}) / (12 \times \text{SCS})] \leq N_{\text{RB}} - 26$.

Specifically:

- The number of RBs for common control resources is specified by **NRDUCellCoreset.CommonCtrlResRbNum**.
- The downlink NARFCN of a cell is specified by **NRDUCell.DlNarfcn**.
- The SCS is specified by **NRDUCell.SubcarrierSpacing**.
- N_{RB} is the number of RBs associated with the cell bandwidth in Table 5.3.2-1 of 3GPP TS 38.104.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-4 and **Table 5-5** describe the parameters used for function activation and optimization, respectively.

Table 5-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
BWP Configuration Switch	NRDUCellBwp.BwpConfigSwitch	UE_BW_ADAPTIVE_SW	It is recommended that this option be selected when there are UEs that do not support the bandwidth of the local cell.
Uplink Minimum Carrier Bandwidth	NRDUCellBwp.UlMinCarrierBw	None	Set this parameter to the minimum uplink bandwidth supported by the network when a cell serves UEs whose allowable bandwidth capabilities do not allow for the bandwidth of the serving cell.
Downlink Minimum Carrier Bandwidth	NRDUCellBwp.DlMinCarrierBw	None	Set this parameter to the minimum downlink bandwidth supported by the network when a cell serves UEs whose allowable bandwidth capabilities do not allow for the bandwidth of the serving cell.

Table 5-5 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
BWP Configuration Switch	NRDUCellBwp.BwpConfigSwitch	UL_SIGNAL_LEAK_OPT_SW	You are advised to contact Huawei engineers to determine whether to select this option. Before selecting this option, ensure that the UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter has been selected.

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	PUCCH_FR EQ_POS_O PT_SW	It is recommended that this option be selected when PUCCH RBs do not experience strong interference. The interference to each uplink RB can be observed using the N.UL.NI.Avg.PRBO to N.UL.NI.Avg.PR272 counters.
Compatibility Algorithm Switch	gNodeBParam.CompatibilityAlgoSwitch	CHANNEL_BW_100M_UE_CAP_R16_SW	It is recommended that this option be selected when the cell bandwidth in n40 is 100 MHz.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD scenario

```
//Enabling UE bandwidth adaptation and configuring the minimum carrier bandwidth  
MOD NRDUCELLBWP: NrDuCellId=0, BwpConfigSwitch=UE_BW_ADAPTIVE_SW-1, UMinCarrierBw=60M,  
DlMinCarrierBw=60M;
```

UE bandwidth adaptation optimization

```
//Enabling uplink signal leakage interference optimization  
MOD NRDUCELLBWP: NrDuCellId=0, BwpConfigSwitch=UL_SIGNAL_LEAK_OPT_SW-1;  
//Enabling PUCCH frequency-domain position optimization  
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=PUCCH_FREQ_POS_OPT_SW-1;
```

Turn on the R16 100MHz Chn Bw UE Capb Use Sw switch when the cell is set up on the n40 frequency band.

```
//Turning on the R16 100MHz Chn Bw UE Capb Use Sw switch  
MOD GNODEBPARAM: CompatibilityAlgoSwitch=CHANNEL_BW_100M_UE_CAP_R16_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling uplink signal leakage interference optimization before disabling UE bandwidth adaptation if  
uplink signal leakage interference optimization has been enabled  
MOD NRDUCELLBWP: NrDuCellId=0, BwpConfigSwitch=UL_SIGNAL_LEAK_OPT_SW-0;  
//Disabling UE bandwidth adaptation  
MOD NRDUCELLBWP: NrDuCellId=0, BwpConfigSwitch=UE_BW_ADAPTIVE_SW-0,  
UlMinCarrierBw=NOT_CONFIG, DlMinCarrierBw=NOT_CONFIG;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Prerequisites

UEs which do not support the cell bandwidth access the network.

Procedure

- Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.
- Step 2** Choose **Signaling Trace Management > Uu Interface Trace (SA networking)/X2 Interface Trace (NSA networking)**.
- Step 3** In SA networking, check whether the value of the *carrierBandwidth* IE in the RRCReconfiguration message in the displayed **Uu Interface Trace** dialog box is consistent with the parameter setting.

In NSA networking, check whether the value of the *carrierBandwidth* IE in the SgNB Addition Request Acknowledge message in the displayed **X2 Interface Trace** dialog box is consistent with the parameter setting.

Perform the following verification operations in SA and NSA networking:

1. In the *ServingCellConfig* IE, check whether the value of **carrierBandwidth** in **SCS-SpecificCarrier** of **uplinkChannelBW-PerSCS-List** is consistent with the **NRDUCellBwp.UlMinCarrierBw** parameter setting. If they are consistent, the **NRDUCellBwp.UlMinCarrierBw** parameter setting is used as the UE effective bandwidth in the uplink.
2. In the *ServingCellConfig* IE, check whether the value of **carrierBandwidth** in **SCS-SpecificCarrier** of **downlinkChannelBW-PerSCS-List** is consistent with the **NRDUCellBwp.DlMinCarrierBw** parameter setting. If they are consistent, the **NRDUCellBwp.DlMinCarrierBw** parameter setting is used as the UE effective bandwidth in the downlink.

In TDD scenarios, if both of the preceding conditions apply, UE bandwidth adaptation has taken effect.

In SUL scenarios, if the first condition applies, UE bandwidth adaptation has taken effect.

----End

5.4.3 Network Monitoring

This function increases the number of cell access attempts and the cell access success rate (in SA networking) and the SgNB addition success rate (in NSA networking), which can be monitored using the following counters:

- SA networking
 - Number of cell access attempts = **N.RRC.SetupReq.Att**
 - Cell access success rate = $\frac{\mathbf{N.QosFlow.Est.Succ}}{(\mathbf{N.QosFlow.Est.Att} - \mathbf{N.QosFlow.Est.Att.EPSFB} - \mathbf{N.QosFlow.Est.Att.EmcFB})} \times 100\%$

- NSA networking

It is recommended that the cell access success rate on the NR side be observed using the SgNB addition success rate:

SgNB addition success rate = $\frac{\mathbf{N.NsaDc.SgNB.Add.Succ}}{\mathbf{N.NsaDc.SgNB.Add.Att}} \times 100\%$

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.101: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *UL and DL Decoupling*
- *Carrier Aggregation*
- *Multi-Frequency Smart Selection*
- *5G Networking and Signaling*
- *Channel Management*
- *UE Power Saving*
- *Cell Management*
- *Super Uplink*
- *MIMO (TDD)*
- *Random Access Control*
- *RedCap*
- *FWA*
- "Technical Specifications" of RF modules in *3900 & 5900 Series Base Station Product Documentation*